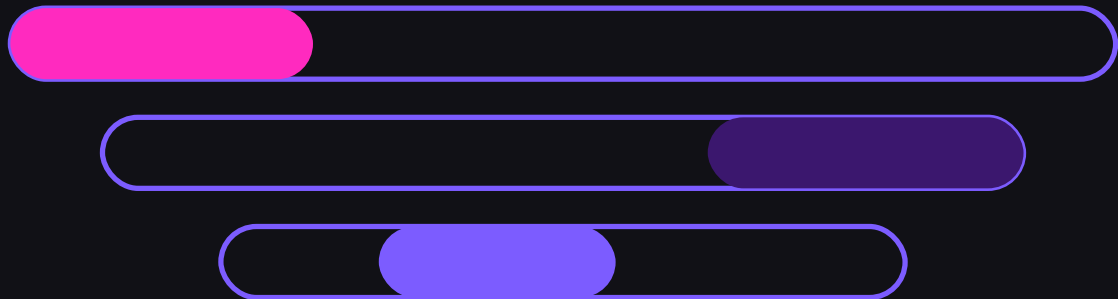


The Body Is Part of the Algorithm

Why the future of Physical AI may depend as much on morphology and materials as models

A designed article on robotics, embodiment and the changing boundary between engineered and living systems.



Why the future of Physical AI may depend as much on morphology and materials as models

The usual story says walking needs a brain.

In 1990, Tad McGeer showed it mostly needs the right geometry. Give a pair of mechanical legs the right proportions, put them on a shallow slope and give them a push. They walk.

No motors drive the joints. No controller calculates balance. Gravity supplies the energy, and geometry, mass distribution and mechanical dynamics produce the gait.^[1]

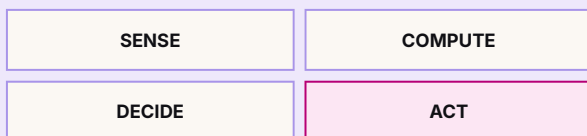
Three decades and several AI revolutions later, the question still stands: how much intelligence does a physical system actually need if its body is designed well?

It matters because every robot acts through a body. The body decides what has to be sensed, how hard the control problem is, how much uncertainty the software must absorb, and which behaviours come for free from mechanics. Physical AI has a hardware question hiding inside its software question.

We inherited a very computational idea of the body

Modern computing suggests a tidy architecture: a processor in the middle, peripherals around it. Cameras supply data, force sensors supply touch, motors move things. Robotics often follows the same pattern.

THE COMPUTATIONAL DEFAULT



Biology spreads the work out. Muscles, tendons, skin, joints and bones shape behaviour continuously. Elasticity stores energy. Softness absorbs impact. Fingers conform to objects. Some sensory processing happens far from the brain.

BRAIN + BODY + ENVIRONMENT

This is the core of morphological computation and physical intelligence.^[2] In practice it means body design can change how much computation a task needs, and what kind.

A useful body doesn't have to be a computer

Careful, though. The language in this field gets loose fast. A spring that stores energy is useful. A soft gripper that wraps around an object is useful. Neither is automatically computing anything.

Vincent Müller and Matej Hoffmann drew the line explicitly. They separate morphology that helps control, morphology that helps perception, and the rarer cases where physical dynamics genuinely perform a computation.^[3]

A compliant gripper usually just makes the problem easier. Picture a rigid hand approaching a slightly irregular object. It needs accurate perception, contact estimation and precise motor control. Swap in compliant fingers and they deform around the object. Some positional uncertainty vanishes into the mechanics, and the object itself helps shape the grasp.

The robot still needs intelligence. The body has simply lowered the bill.

Same AI, different body

Machine learning makes this sharper. In 2026, researchers compared robotic grippers on decluttering tasks using deep reinforcement learning. One design, the Soft ScoopGripper, had a scoop-like structure that naturally supported pushing while grasping. That was enough for the learned policy to clear objects without separately learning to push and to grasp.^[4]

Give the same learning architecture a different morphology and you change the learning problem.

So future robot benchmarks may need to ask two questions at once: how capable is the model, and how much difficulty did the body create for it? A leaderboard that ignores the gripper is partly grading the gripper.

Sometimes the body really does compute

There are cases where the stronger claim holds. In 2015, Kohei Nakajima and colleagues put a soft silicone arm in water and used its nonlinear dynamics and short-term memory as a physical reservoir computer. [\[5\]](#)

Soft bodies are hard to control: they bend in complex ways, have many coupled degrees of freedom, and retain traces of earlier states. The researchers used exactly those properties as a resource. Inputs changed the arm's state, its dynamics transformed them, and embedded sensors read the result. The body was part of the information processing.

Logic can live in matter

Once computation leaves silicon, the line between body and controller gets messy.

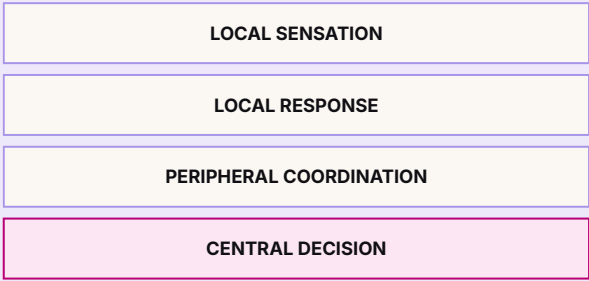
In 2016, Harvard's Octobot showed a fully soft autonomous robot whose movement was sequenced by microfluidic logic rather than conventional rigid electronics. [\[6\]](#) In 2024, researchers built integrated mechanical computing from chains of bistable elements, where mechanical waves carried information, performed logic operations and triggered mechanical output directly. [\[7\]](#)

Their computational power is tiny next to a digital processor. What matters is where they sit: at the events robots meet constantly, such as contact, slip, deformation, pressure and impact. Sending every one of those events through a central AI model may simply be poor system design.

Intelligence moves toward the edge of the body

The octopus is a useful model because its nervous system is so distributed. The 2026 octopus-inspired arm from Article 1 has ten sensorized suction cups. Individual cups detect contact and trigger local suction, while higher-level electronic control coordinates the whole arm. [\[8\]](#)

DISTRIBUTED CONTROL



Higher-level control no longer has to process every tactile event in detail. Expect this kind of architecture in messy, contact-rich environments. Some decisions need planning. Some need perception. Some need reflex. And some are best left to physics.

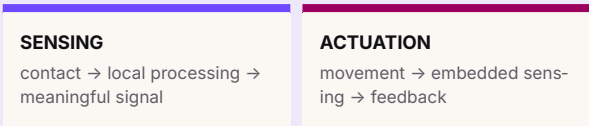
Sensing and actuation start to blur

Conventional sensors produce data: pixels, numbers, interpretation later. Biological sensing does more of its work at the periphery.

In 2024, researchers demonstrated an artificial organic afferent nerve that combined mechanoreceptors with a synaptic transistor. It integrated tactile information, encoded it in neural-like spikes and supported fast slip detection in a robotic gripper. [\[9\]](#) The information flows differently: sensor plus local processing produces a meaningful signal, and only then does it reach higher-level control. That can cut latency, bandwidth and computational load.

Actuators can do double duty too. An artificial muscle-tendon system combines liquid-crystal-elastomer actuators with embedded liquid-metal channels. The same structure contracts and reports its own force and length, enabling closed-loop control in robotic fingers and grippers. [\[10\]](#)

SENSING AND ACTUATION START TO BLUR



Even memory doesn't have to be digital. Mechanical neural networks built from lattices of tunable beams can be trained toward desired physical behaviours by changing the stiffness of their elements.^[11] A 2024 material system integrated sensing, mechanical memory, sequential logic and actuation.^[12] None of this is close to general-purpose AI. It does show that some information can live in physical properties themselves.

An intelligence budget

I find it useful to think of every robot as having an intelligence budget.

Every physical task creates problems that must be solved somewhere: stability, contact, slip, grasping, uncertainty, state estimation, planning, damage. Traditional robotics gives most of them to sensors, software and control. A more distributed design can spread them across geometry, mechanics, materials, sensors, local reflexes, physical memory, learned models and high-level reasoning.

So the design question is about placement. Where is the fastest, cheapest and most robust place to solve each problem?

A language instruction belongs in the AI system. A contact reflex probably doesn't. A small geometric uncertainty may be easier to absorb through compliance than through another layer of perception. And a material can absorb an impact before any processor has time to react.

AN INTELLIGENCE BUDGET	
CLOSER TO PHYSICS	CLOSER TO ABSTRACTION
GEOMETRY	shape the problem
MATERIALS	absorb uncertainty
LOCAL REFLEXES	react fast
LEARNED MODELS	adapt & perceive
REASONING	plan & choose

Bigger models don't make bodies disappear

As AI improves, it is tempting to expect hardware to matter less. Software can compensate for poor mechanics to a surprising degree. Learning adapts, perception copes with uncertainty, control stabilises awkward bodies.

The reverse holds just as well. A well-designed body makes learning easier. The ScoopGripper simplified the behavioural problem. The octopus-inspired arm reduced central processing. Passive walking showed decades ago how much behaviour mechanics can deliver alone.

So the real design space is co-optimisation. What should this system learn? And what body makes that easier to learn? Those questions belong together.

The body is part of the algorithm

The title is deliberately provocative, and it needs a caveat. Sometimes the body just makes a problem easier. Sometimes it transforms information. Sometimes materials store state. Sometimes peripheral systems react before the central AI is involved. And sometimes physical dynamics genuinely become part of the computation.

In every case, the intelligence of a physical system can't be judged apart from the body it acts through. Software abstracts away from hardware remarkably well. Physical AI can't abstract away from gravity, friction, elasticity, geometry, contact and material fatigue. The environment pushes back.

Which raises the next question. Biology already has actuators that remodel, sensors that adapt, tissues that repair and structures that build themselves. What happens when those become engineering materials?

When someone tells you a robot is smart, how much of that is the model, and how much is the body?

Next: what changes when the components of a robot are alive.

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